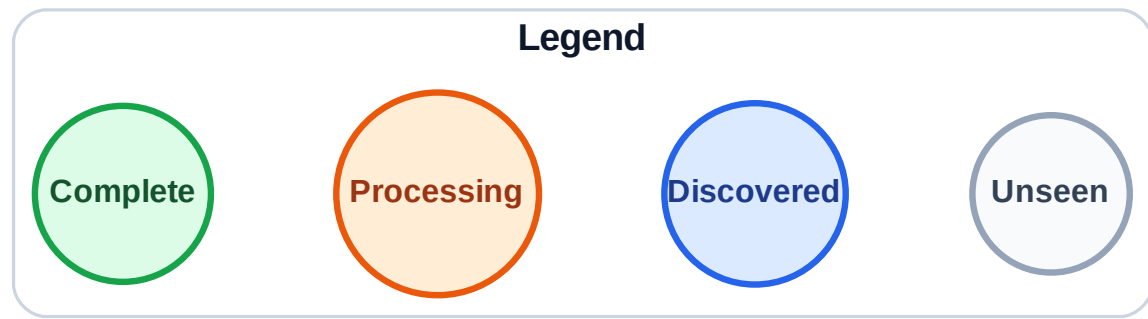


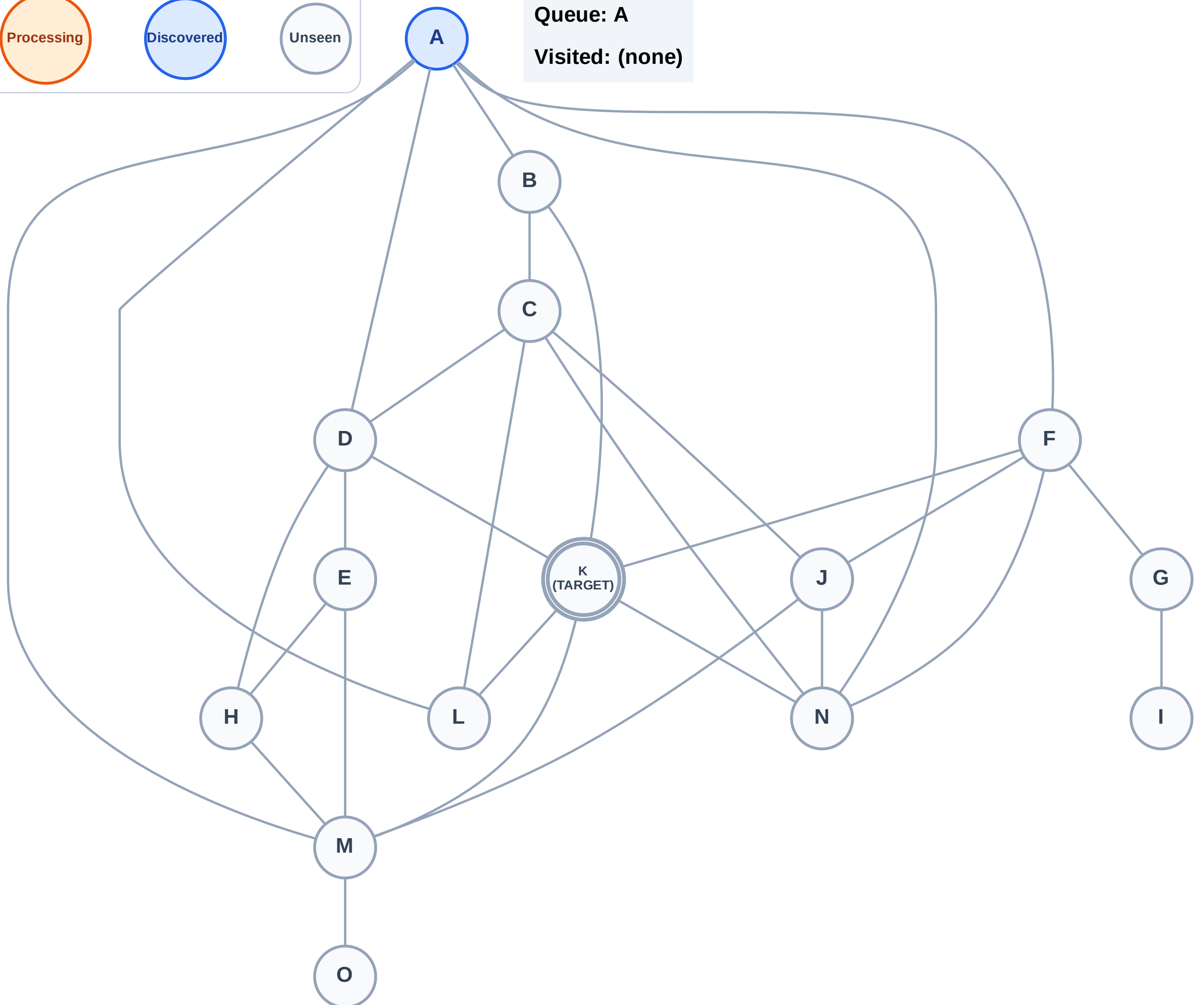
BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))



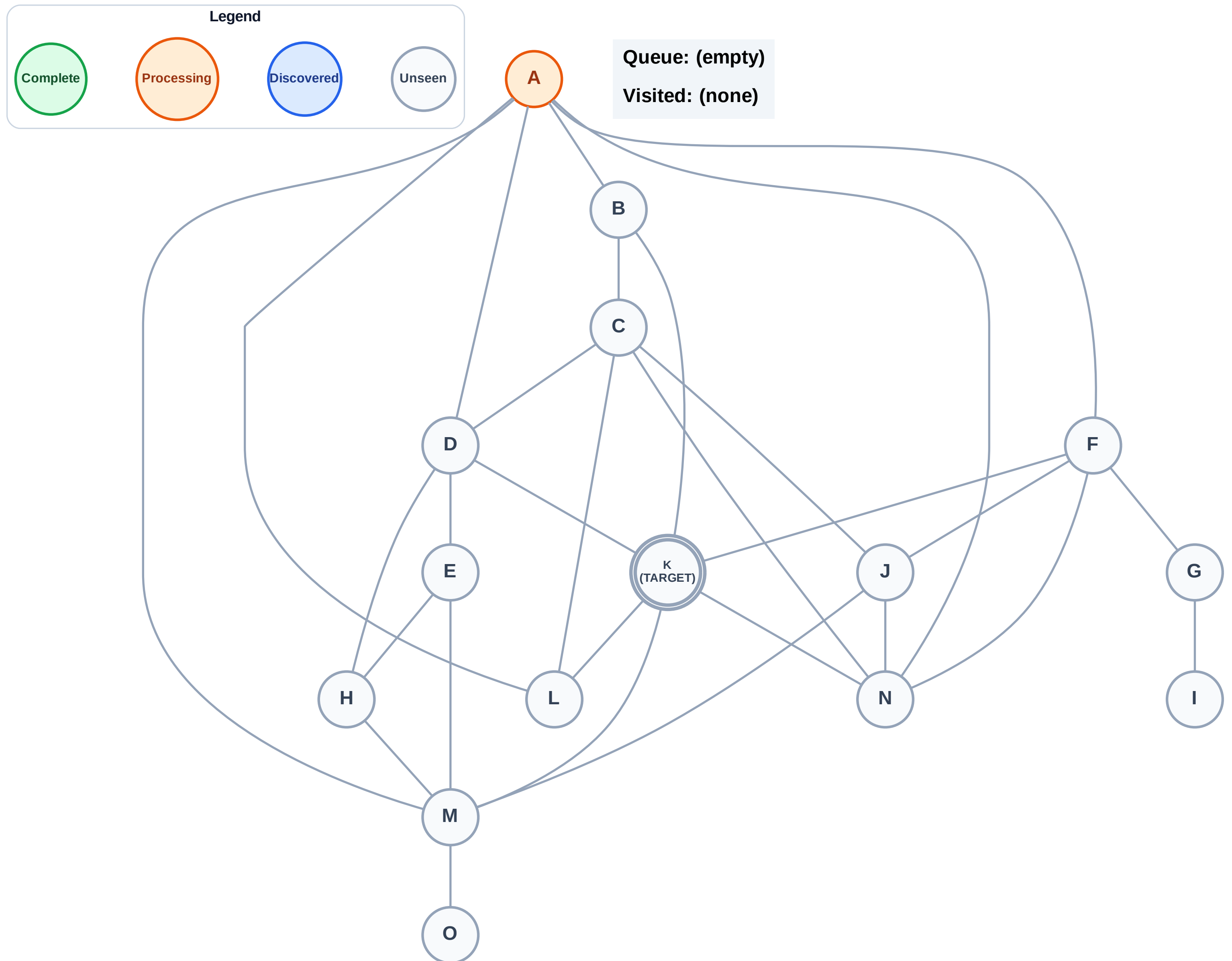
Queue: A

Visited: (none)



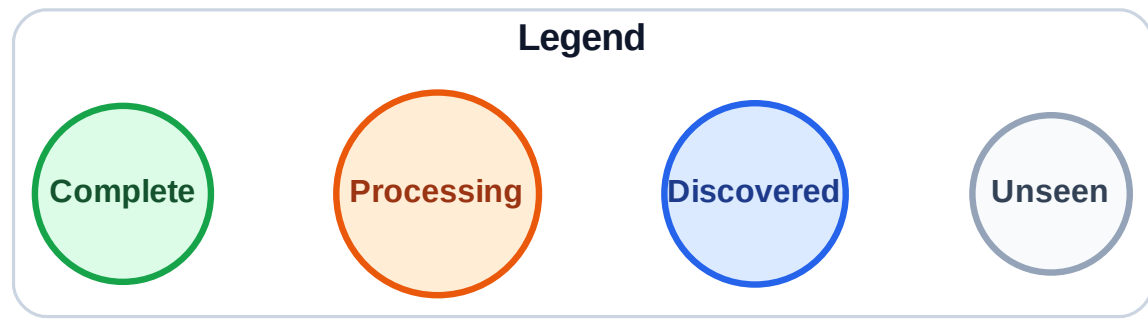
Step 2: Process node A

Step 2: Process node A



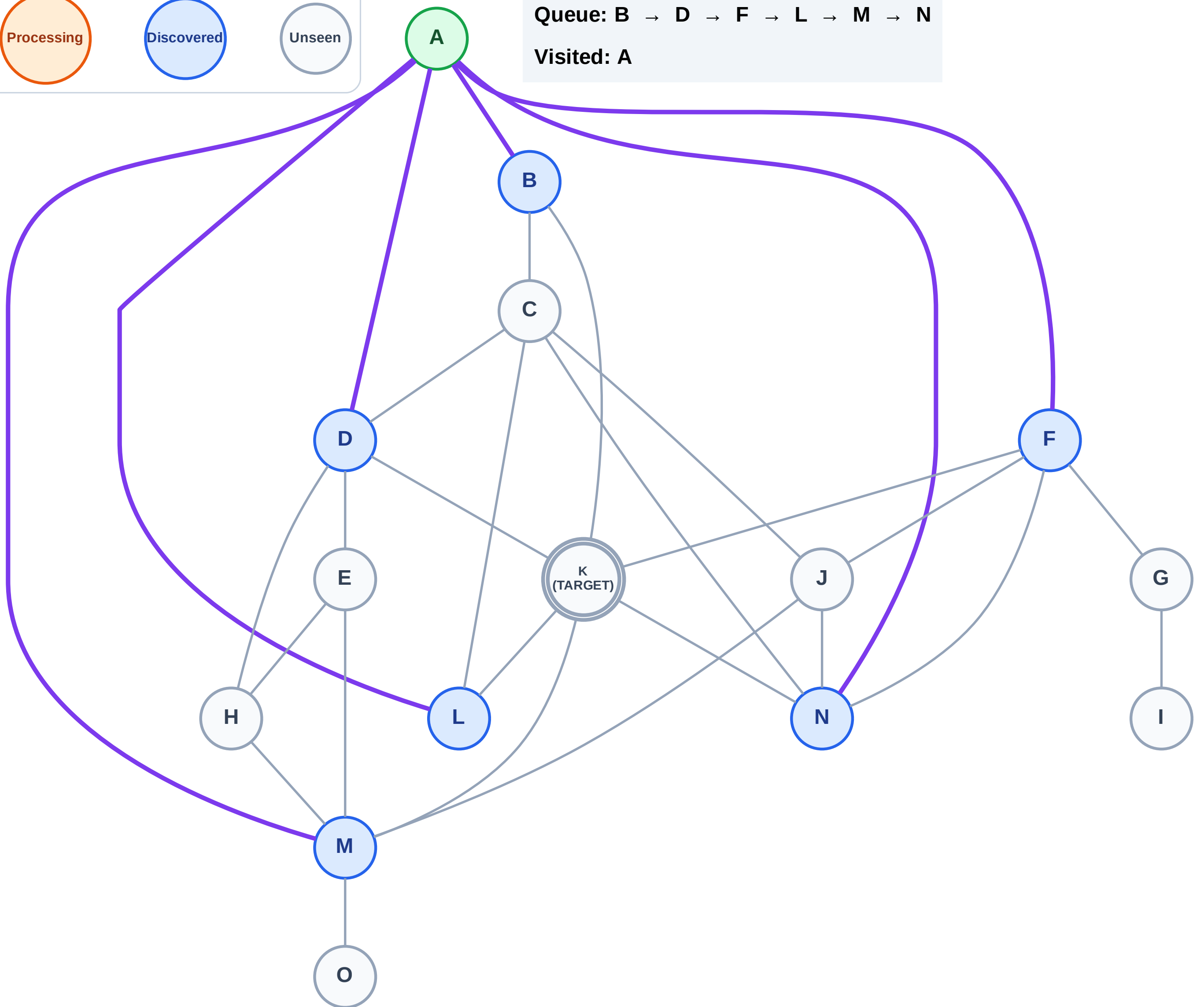
BFS Traversal

Step 3: Discover B, D, F, L, M, N from A



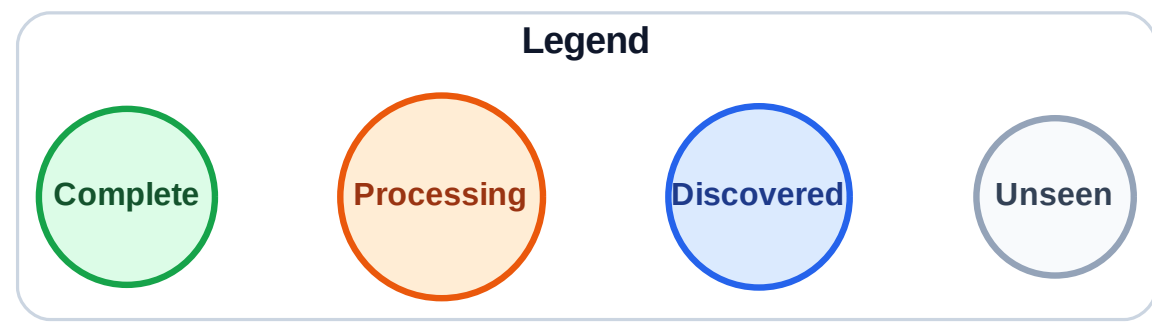
Queue: B → D → F → L → M → N

Visited: A



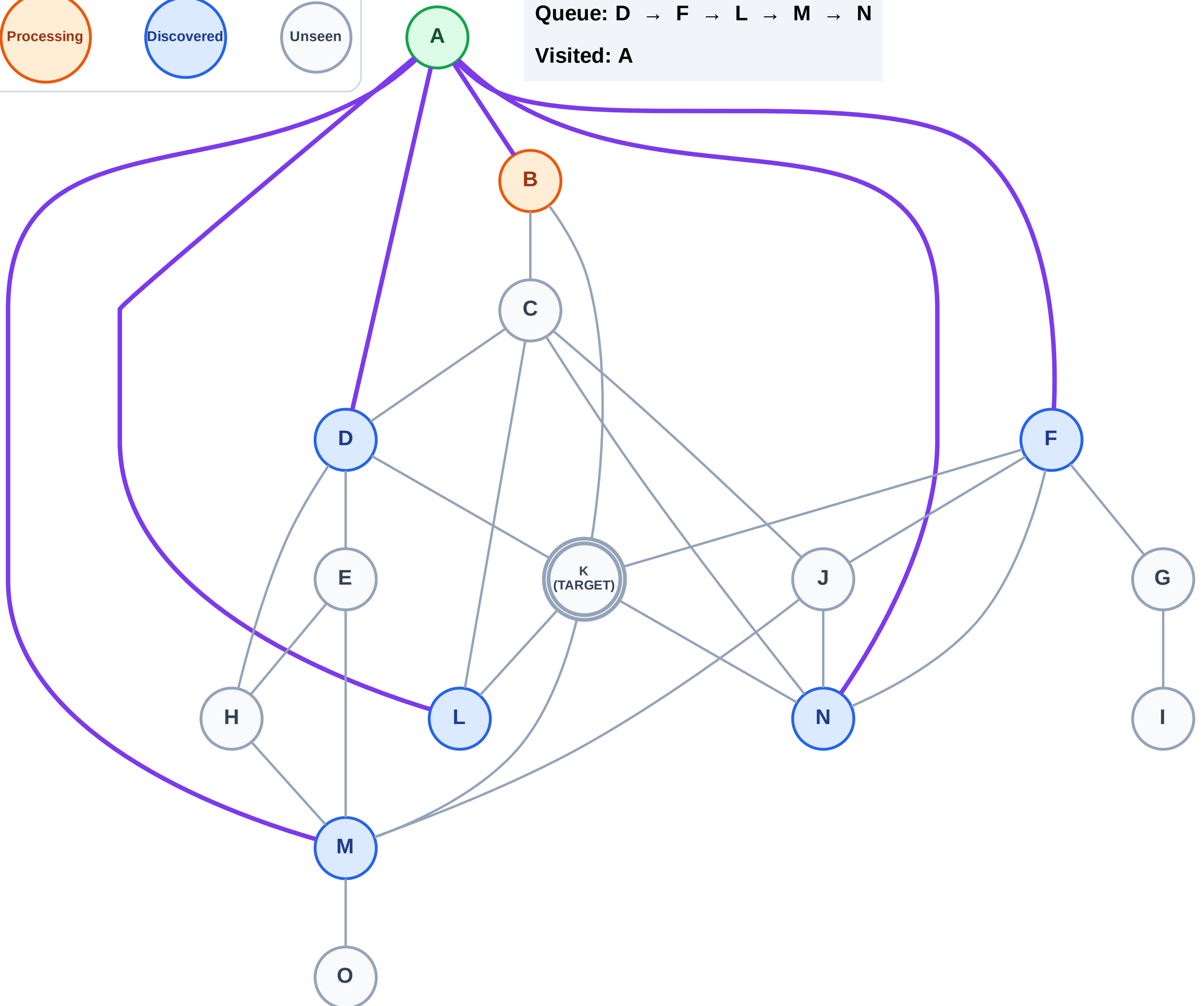
BFS Traversal

Step 4: Process node B



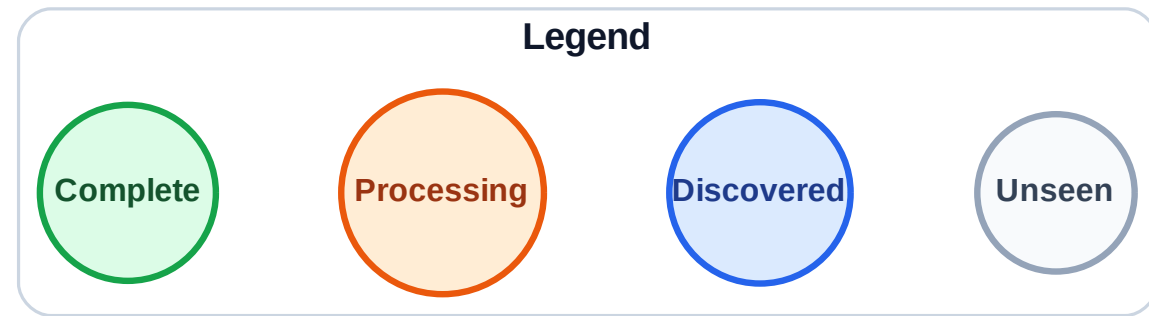
Queue: D → F → L → M → N

Visited: A

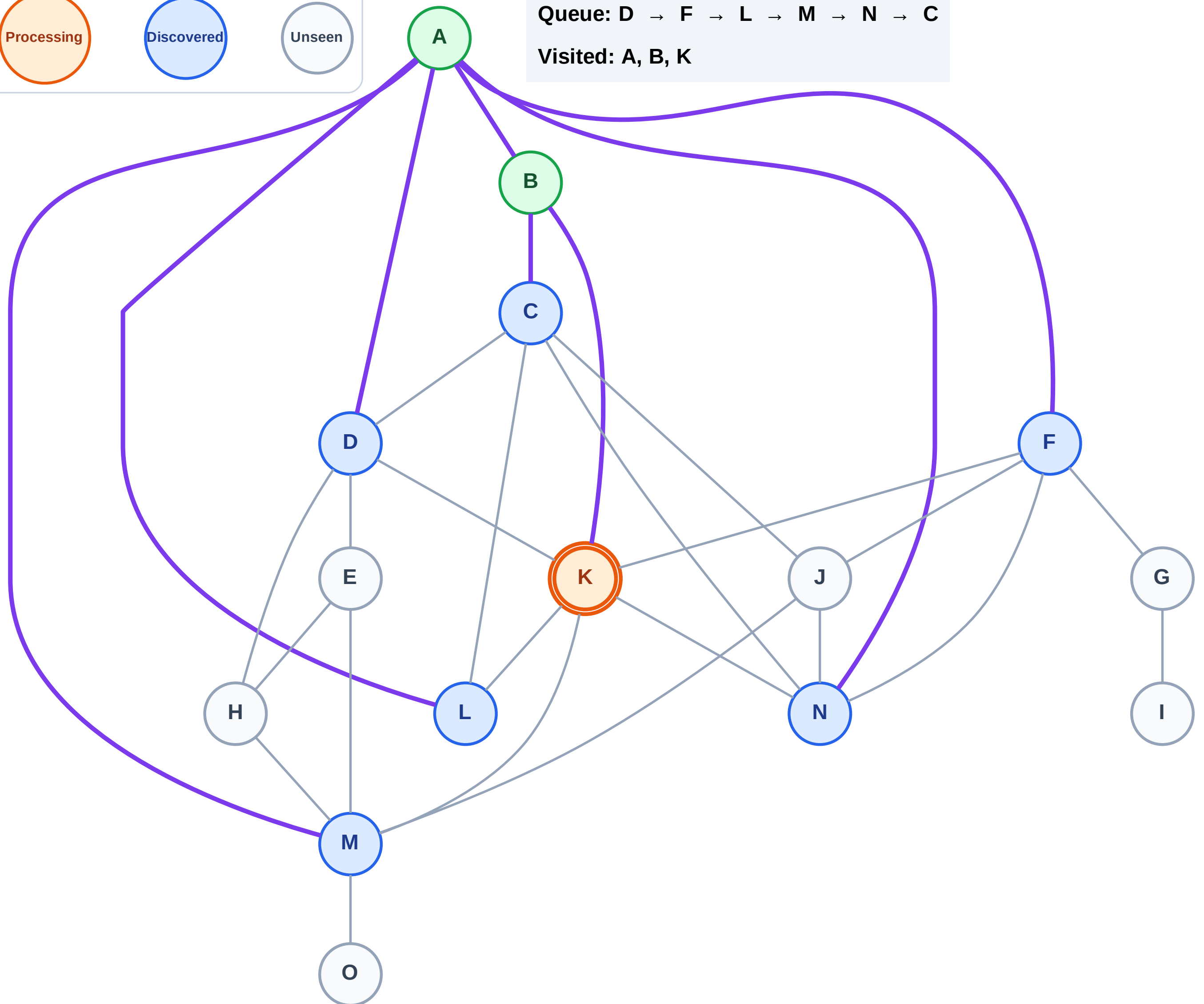


BFS Traversal

Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → B → K

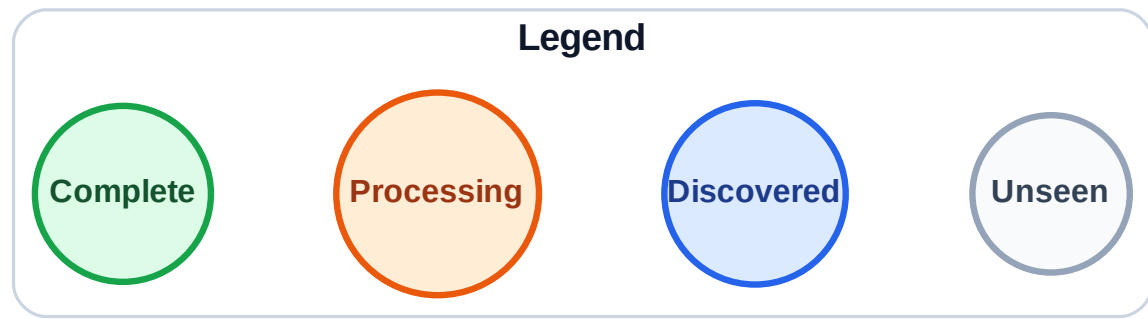


Queue: D → F → L → M → N → C
Visited: A, B, K



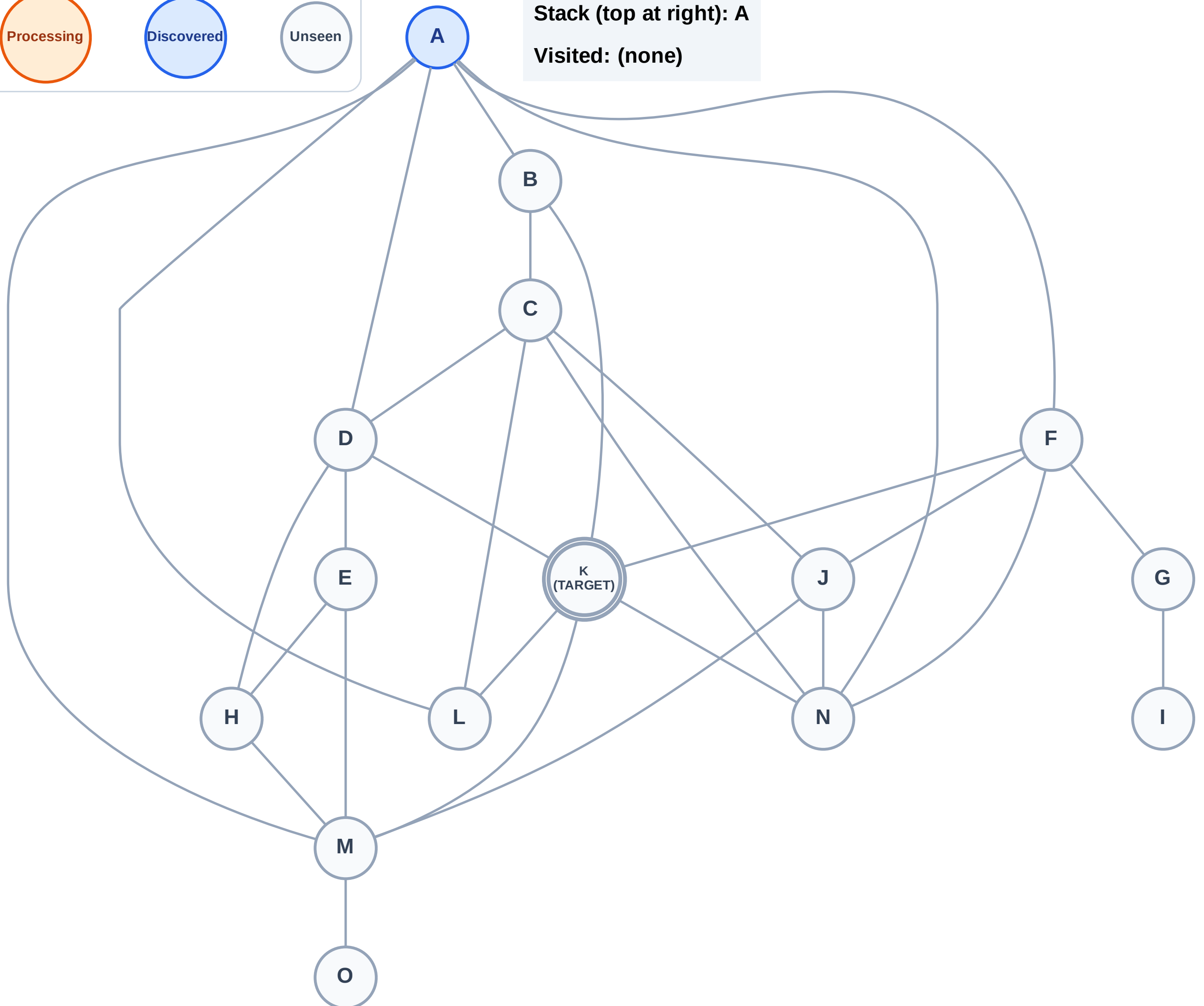
DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

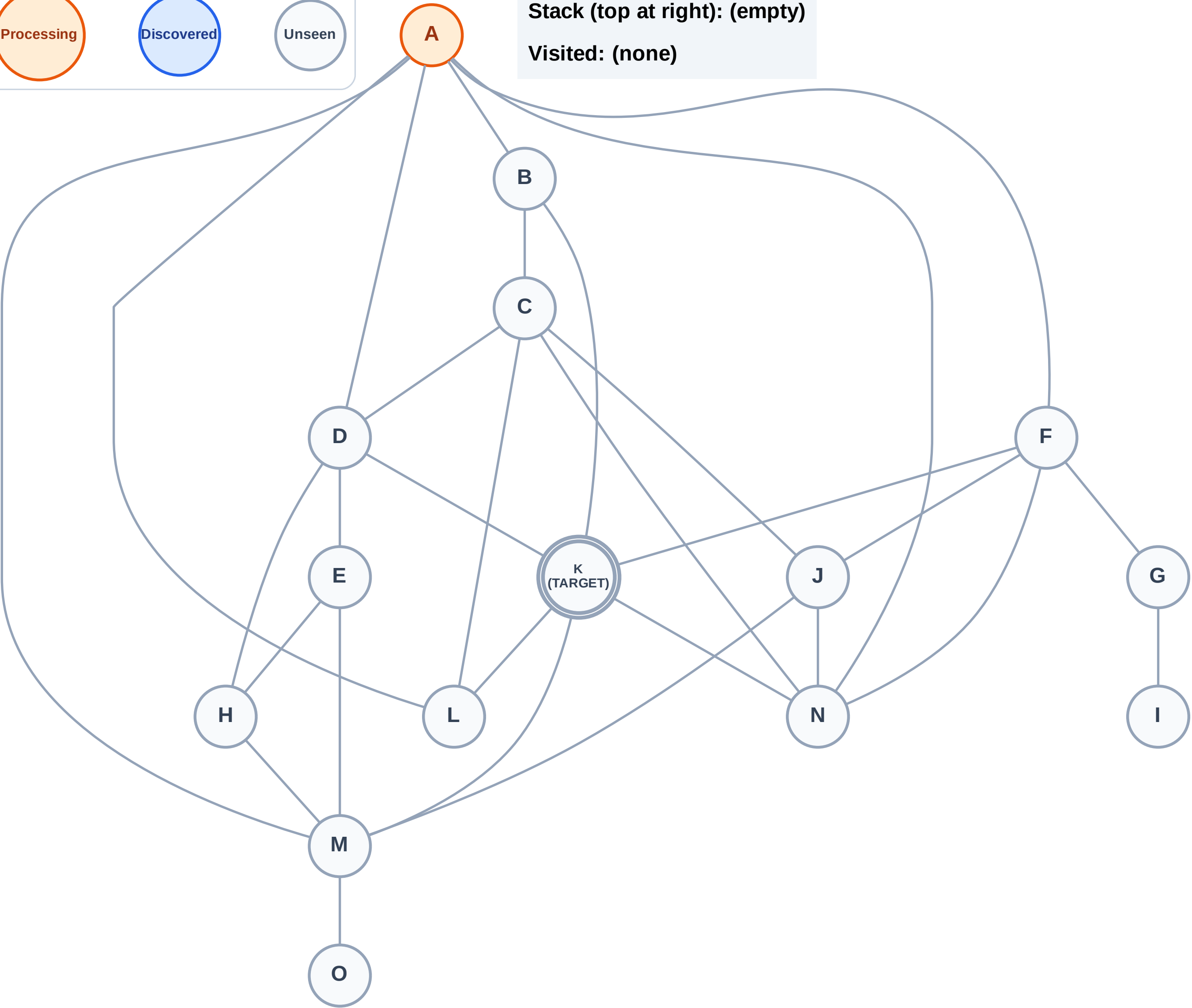
Complete

Processing

Discovered

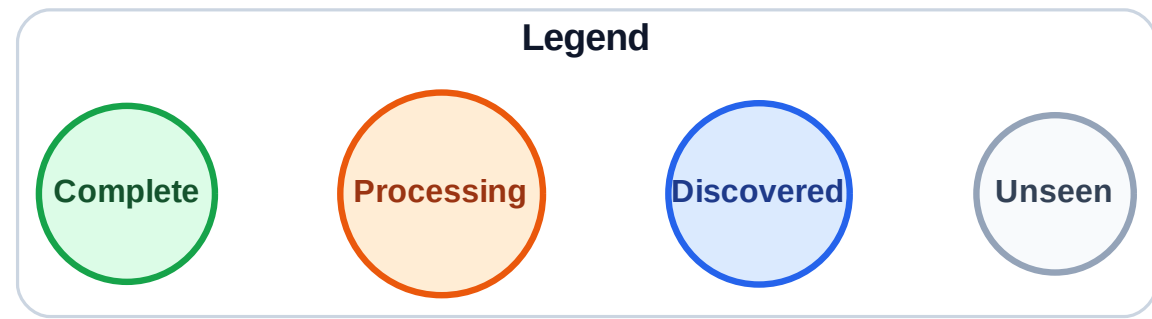
Unseen

Stack (top at right): (empty)
Visited: (none)



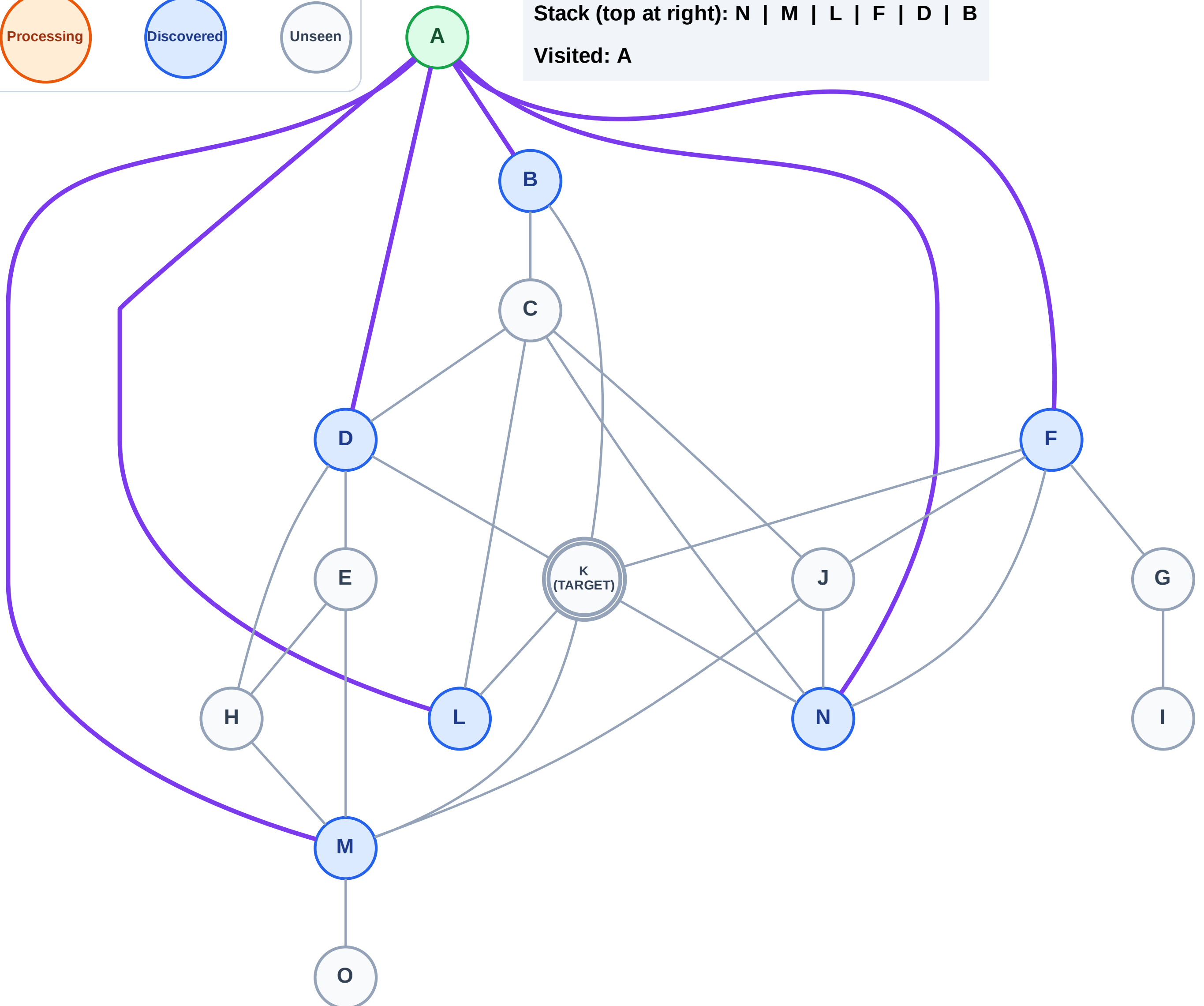
DFS Traversal

Step 3: Discover N, M, L, F, D, B from A



Stack (top at right): N | M | L | F | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

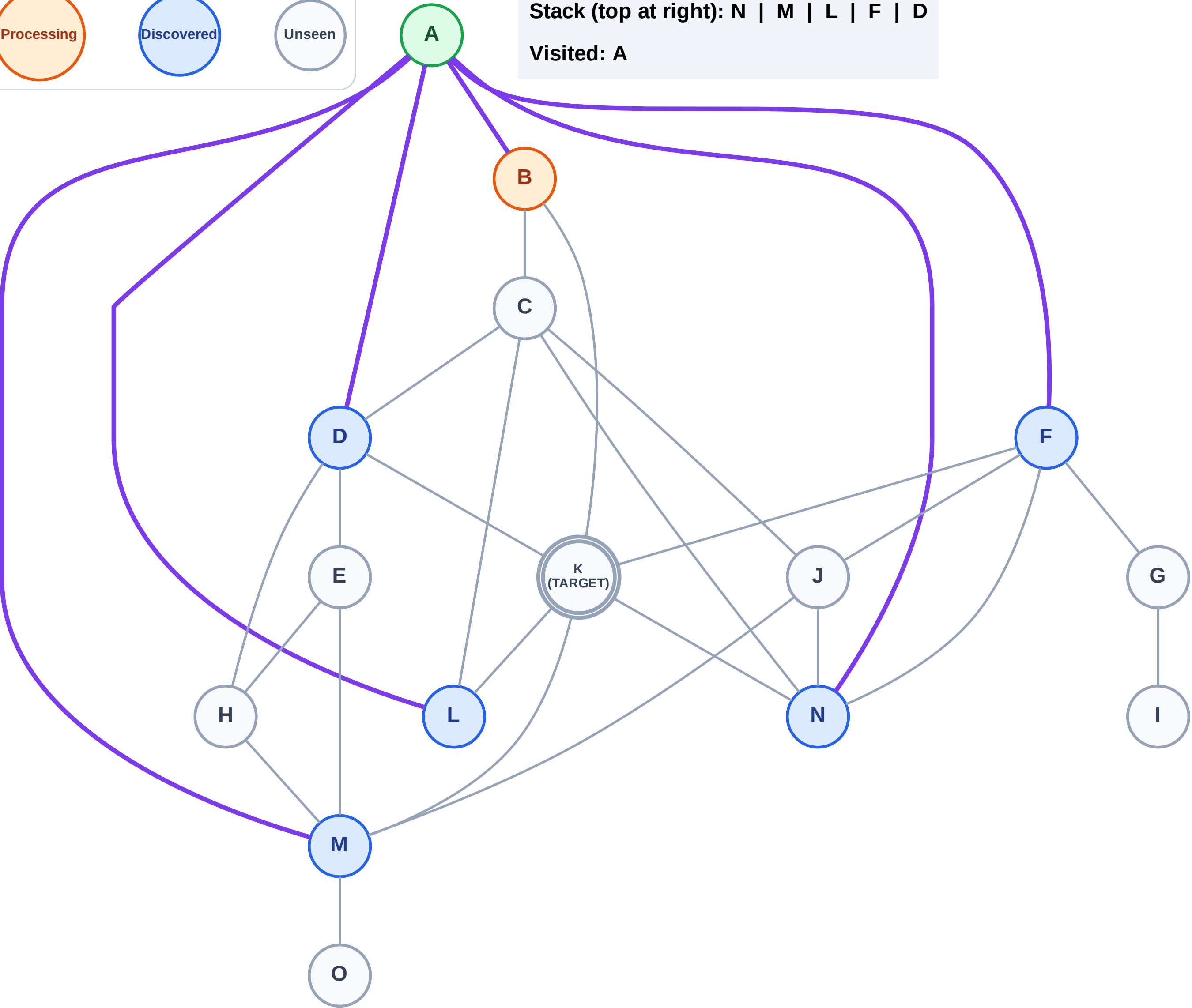
Processing

Discovered

Unseen

Stack (top at right): N | M | L | F | D

Visited: A



DFS Traversal

Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → B → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): N | M | L | F | D
Visited: A, B, K

